

advances have paved the way for the revolution of fields such as quantum physics, artificial intelligence and nanotechnology. These advances will have a profound effect on the way we live and work, the virtual reality we see in movies like the Matrix, may actually come true in the next decade or so.

Let us take a peek into what the future might hold for our computing devices.

1. Nanocomputers

Nanocomputer is the logical name for a computer smaller than the microcomputer, which is smaller than the minicomputer. Technically speaking, it is a computer whose fundamental parts are no bigger than a few nanos. For comparison, the smallest part of current state-of-the-art microprocessors measures 45 nm as of February 21, 2007. But this is all set to change very soon.

1st November 2011. According to a group of researchers calling themselves the Nanocomputer Dream Team, that's the day they'll unveil a revolutionary kind of computer, the most powerful ever seen. The team includes Bill Spence, publisher of Nanotechnology magazine and over 300 enthusiasts with diverse backgrounds - engineers, physicists, chemists, programmers and artificial intelligence researchers.

The most promising idea is rod logic, invented by nanotechnology pioneer Eric Drexler, now chairman of the leading nano think tank The Foresight Institute. Rod logic uses stiff rods made from short chains of carbon atoms. Around each rod sits a knob made of a ring of atoms.

The rods are fitted into an interlocking lattice, where each rod can slide between two positions, and be reset by a spring made of another few atoms. This arrangement has already been used to achieve the effect of a conventional electronic transistor, where the flow of current in one wire is switched on and off by current in a different wire. Once you have



The future of computers

transistors, you can build a NAND gate. From NAND gates you can construct every other logic element a computer needs.

1.1. Electronic Nanocomputers

Given our experience of working with electronic computing devices, it is highly likely advances in nanocomputing technology are likely to come from this direction, thus making the electronic nanocomputers the first ones to come out. Electronic nanocomputers would operate in a manner similar to the way present-day